



An overview of measurements, data compilations and prediction methods for the critical heat flux in water-cooled tubes



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ARTICLE INFO

Keywords:

Critical heat flux
Prediction methods
Experimental methods
Datasets

ABSTRACT

This paper describes the history and state of the art of measuring and predicting the Critical Heat Flux (CHF) in water-cooled tubes. Over the past 70 years, over 40,000 tube-CHF measurements have been obtained in over 160 separate studies. About 17 separate compilations of CHF datasets in water-cooled tubes have been produced, many overlapping each other. In addition, hundreds of tube-based CHF prediction methods can be found in the literature. The pertinent experimental details and concerns for these datasets and the proliferation of CHF prediction methods are discussed. A graphical presentation of the ranges of conditions covered by these CHF datasets is included.

1. Introduction

Before the advent of nuclear reactors, there was no urgent need for CHF measurements as most boiling processes where CHF was encountered were temperature-controlled (e.g., heat exchanger tubes in fossil-fuelled boilers). However, water-cooled nuclear reactors, which are limited in power by the CHF occurrence, are basically heat-flux-controlled systems. Hence, exceeding the CHF can have serious consequences, in particular for Pressurized Water Reactors (PWRs). For this reason, most of the countries with an interest in nuclear energy became active in CHF measurements by the mid 1900s, and have continued doing so up to the present. In the USA, [McAdams et al. \(1949\)](#), and [Jens and Lottes \(1951\)](#) were the first to report flow boiling CHF measurements.

Along with CHF measurements, which were originally limited to tube geometries, CHF prediction methods were also proposed starting in the late 1940s. The earliest tube-CHF prediction methods were primarily empirical. These crude empirical correlations lacked any physical basis, and had a limited range of application. Subsequently, a large number of phenomenological equations or physical models for CHF were developed; some of these models have been used in reactor safety analysis codes. Physical models and phenomenological equations, however, depend on the mechanisms controlling the CHF, which changes with flow regime. This necessitates the use of a combination of different models, equations or correlations for predicting the CHF for the wide range of conditions that can be encountered during reactor

accident transients. Because of this, and because of the large proliferation of CHF equations and correlations (over 500 CHF correlations are currently available for uniformly-heated water-cooled tubes), a more universal CHF prediction methodology was required. Hence, look-up tables for predicting the CHF in water-cooled tubes were subsequently derived ([Groeneveld et al., 1986, 1996, 2007a](#)).

The CHF look-up table is basically a normalized CHF databank for water-cooled tubes. Compared to other available prediction methods, the look-up table approach has the following advantages: (i) greater accuracy, (ii) wider range of application, and (iii) correct asymptotic trend. Although look-up tables were initially developed for tubes, and have been successfully used in subchannel codes, the greatest potential for their application is in predicting the consequences of postulated loss-of-coolant accidents (LOCAs).

It should be emphasized that tube based CHF prediction methods discussed in this paper will generally be of limited usefulness in predicting the CHF in bundle geometries for conditions where bundle CHF data are available. However bundle CHF measurements at accident conditions are generally unavailable, in which case the analyst may employ reliable tube-based CHF prediction methods after correcting for effects such as speed of the transient, axial and radial flux distribution, geometry, spacers etc. Possible expressions to account for these effects have been proposed by [Groeneveld et al. \(2003\)](#), [Tong and Weisman \(1996\)](#) and [Collier and Thome \(1994\)](#).

Note that virtual all bundle CHF measurements were obtained on electrically-heated fuel-bundle simulators, and are subsequently

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extrapolated to in-reactor conditions. Few in-reactor CHF measurements were obtained in the 1960s and 1970s; they have been recently reviewed by Groeneveld (2017) who also presented some concerns related to extrapolating CHF measurements from out-reactor fuel simulators to in-reactor conditions.

Section 2 of this paper discusses the types of CHF experiments and CHF detection methods employed during the past 70 years, as well as various factors that could affect the uncertainty in the CHF measurements. Section 3 describes the experimental datasets and data compilations for CHF measurements in water-cooled tubes. Section 4 provides an overview and the evolution of CHF prediction methods in simple geometries. Section 5 presents graphically the database coverage of the present tube-CHF database and discusses the lack of data at certain flow conditions.

2. CHF measurements

2.1. CHF detection methods

CHF is typically characterized by a noticeable increase in surface temperature in response to a small change in heat flux. This change in temperature can be very drastic, as is the case for PWR-type conditions (referred to as “fast dryout”; see Groeneveld, 1986), or this change can be gradual as for Boiling Water Reactor (BWR) conditions (referred to as “slow dryout”). Over the past 70 years, CHF experimenters have used many different CHF detection methods; the four main methods are summarized below.

Visual The early CHF papers identified the CHF as the heat flux at which the test section “started to redden visually”. This method was used by some of the early researchers (e.g., Hood and Isakoff, 1962). Although this method could work well for fast dryouts or Departure from Nucleate Boiling (DNB) at subcooled CHF conditions, the slow dryouts, typical for BWR conditions, would only result in modest temperature excursions that did not result in a discoloration of the heated surface.

Physical Burnout At high flow rates and high subcoolings, the CHF is typically very high, making it difficult to quickly switch off the power at CHF to avoid test section failure. Several investigators reported that their CHF corresponded frequently to a physical burnout at CHF occurrence (e.g., Hood and Isakoff, 1962; Pabisz and Bergles, 1996).

Change in Test Section Resistance The test section material used in most CHF experiments is either Inconel or stainless steel. Inconel has a very low temperature coefficient of resistivity compared to stainless steel which has a much higher value. By using a stainless steel test section as one leg of a Wheatstone bridge, CHF can be detected when the change in test section resistance due to a significant temperature excursion results in an imbalance in the Wheatstone bridge, triggering a power supply trip. This method of CHF detection was reported by Dell et al. (1969), Matzner et al. (1965), Hewitt et al. (1965) and others.

Test Section Thermocouples The most common method for detecting CHF is using thermocouples attached to the downstream end of the heated length of the test section. This method is very effective for most types of CHF occurrences, with the possible exception of very fast temperature excursions where a method based on detecting a change in test section resistance may be more reliable. For very slow dryouts, the thermocouple method may not always be effective because of the absence of a noticeable dryout temperature excursion. Here a more reliable method is based on monitoring the change in the slope of heated wall temperature vs. heat flux or $\Delta T_w/\Delta q$ (Groeneveld, 1986).

Other CHF detection methods that have been explored include those based on acoustic, ultrasonic and infrared techniques; these novel methods however are not yet sufficiently mature and have not been employed in obtaining the more than 160 tube-CHF datasets used for deriving CHF prediction methods.

In some cases, the CHF was actually a “by-product” of a film-boiling experiment during which detailed wall temperature distributions were

measured. For any given heat flux, the CHF quality was either assumed to be the quality where the first rise in surface temperature was detected, or defined as the average of the last pre-CHF quality and the (subsequent) first post-CHF quality. Examples of this type of CHF measurement are given by Era et al. (1966), Bennett et al. (1967) and Herkenrath and Mork-Morkenstein (1969).

2.2. Primary and secondary CHF parameters

CHF is primarily a function of flow conditions and test section geometry. Since the look-up table and the overwhelming majority of empirical tube-CHF correlations are based only on CHF measurements obtained in a tubes having a uniform axial flux distribution, the impact of non-circular geometries and axial flux distribution on CHF will not be discussed.

During CHF experiments, the CHF is a function of the following primary parameters: pressure (either at the start of the heated length or at the CHF location), inlet temperature, mass flow rate, diameter (D) and heated length (L). It has been shown by several experimenters (e.g., Lee and Obertelli, 1963; Lee, 1965) that the primary parameters heated length and inlet temperature can be replaced by thermodynamic quality at the CHF location, provided that the heated length is sufficiently long (e.g., $L/D > 50$) to remove any upstream history effects. Thus, for a given inside diameter, CHF becomes a function of the flow conditions at CHF, i.e., mass flux, pressure, and thermodynamic quality. These are the three parameters of the CHF look-up table.

The following secondary parameters could also affect the CHF in uniformly heated tubes:

- **Test section orientation:** Although most CHF tests have been performed for upward flow in vertical test sections, some investigators have investigated the CHF behavior in horizontal flow and down-flow (e.g., Wong et al., 1990). Based on an extensive analysis of CHF in horizontal tubes, Wong has shown that the effect of test section orientation is not significant at high mass velocities where flow stratification is suppressed. The boundaries of flow stratification can be estimated from flow regime maps such as those proposed by Taitel and Dukler (1975).
- **Test section material:** Test section material in general has little effect on CHF during flow boiling. However, for low flows and conditions where CHF is due to DNB, highly conductive test section materials could dissipate hot spots under bubbles and thus increase the CHF (Berenson, 1962).
- **Type of heating:** The large majority of CHF experiments are performed on directly heated tubes (Joule heating), while the most typical application is indirect heating of a fuel sheath where the heat source is nuclear heat. Leung et al. (1982) compared experimentally the CHF for direct and indirect heating and observed no significant difference.
- **Wall thickness:** Several experimenters (e.g., Bergles, 1963; Bennett et al., 1965) investigated the wall thickness effect of CHF but found no discernible effect. Some effect could possibly be present for very thin walls that could limit the heat diffusion of hot spots during a DNB-type CHF.
- **Surface roughness:** Most test sections have a very smooth surface finish (similar to a fuel sheath) and the impact of the very small surface roughness is not found to be significant. Even for cases with a machined surface roughness, the impact of roughness on CHF is generally small as it is usually the vapor generation rate at the surface that determines the CHF occurrence. However when the surface roughness becomes larger than the film thickness (in annular film dryout), a reduction in CHF could occur due to premature film breakdown. In addition, a roughened surface provides preferential nucleation sites that could affect the CHF when DNB is the main mechanism of CHF occurrence (Bergles, 1976; Berenson, 1962).
- **Inlet /outlet throttling:** In the majority of the CHF experiments, the

flow was considered free from fluctuations. However, if the flow were restricted at the outlet, an instability in flow and pressure could occur, resulting in significant reductions in CHF as was observed by Lowdermilk et al. (1958) and Mayinger et al. (1966). Kirillov (1997) also observed flow oscillations with a “soft” inlet (no throttling), as opposed to a “hard” inlet (with throttling) that generally suppressed the occurrence of flow oscillations.

- Dissolved gas content in the coolant: Most CHF experiments used degassed water, but some have purposely used dissolved gas in the coolant to examine its effect on boiling heat transfer. Kirillov (1997) reported on Russian experiments where reductions of up to 30% in CHF were observed for dissolved gas content of 4000 normal cm³/kg.

2.3. CHF reproducibility

In the late 1960s and 1970s, the concern was raised in Europe that there could be significant differences in CHF between measurements obtained at different heat transfer laboratories on nominally the same test section and for the same flow conditions. Therefore, two independent reproducibility exercises were performed, one in Western Europe in 1970 (CISE, 1970; Nilsson, 1970) and one reported in the Russian literature (Kirillov et al., 1985; Kirillov, 1997). The objective of these reproducibility studies was to determine the variation in CHF measurements between various laboratories. Eight to ten laboratories participated in each of these two reproducibility studies. In general, the agreement between the various European laboratories was within 10% except for two outliers that were later disqualified. Details of these reproducibility studies are provided in Appendix A.

3. Tube-CHF database

3.1. CHF datasets

In the USA, McAdams et al. (1949) and Jens and Lottes (1951) were the first to report flow boiling CHF measurements in simple geometries such as electrically heated tubes, annuli or rectangular channels (see also an early review of CHF studies by DeBortoli et al., 1958). A large expansion in CHF experimentation took place in the 1960s and 1970s in parallel with the world-wide construction of water-cooled nuclear reactors. A recent NUREG report (Groeneveld, 2017) identified over 160 CHF experimental datasets obtained in water-cooled tubes. In the large majority of these experimental studies, CHF was detected by a temperature excursion measured from thermocouples located near the downstream end of the heated length.

3.2. CHF data compilations

Analyzing multiple CHF datasets is a cumbersome task, so many CHF investigators created CHF data compilations just for water-cooled tubes. One of the earliest CHF databases available was that of DeBortoli et al. (1958), who collected over 1000 tube-CHF data. Thompson and Macbeth (1964) reported their compilation of tube CHF data containing over 4000 points obtained mainly from laboratories in the USA and Europe. Groeneveld et al. (1986) described a large CHF database having over 15,000 points; this compilation was a result of an extensive collaboration between Atomic Energy of Canada Ltd and the University of Ottawa (and is called the “AECL-UO CHF database”). In parallel, Kirillov and coworkers at the Institute of Physics and Power Engineering (IPPE) in Obninsk compiled a CHF database of over 14,000 points from sources primarily from the Union of Soviet Socialist Republics (USSR). Subsequently, Groeneveld and Kirillov combined their databases expanding the total number of data points beyond 29,000; the combined database is referred as the “AECL-UO-IPPE CHF database” (Groeneveld et al., 1996). In parallel, Hall (1999) and Hall and Mudawar (2000), compiled a database of over 30,000 tube CHF data, which covered

tubes of very small diameters ($D < 2$ mm) that were not included in the AECL-UO-IPPE tube database. The latest CHF look-up table (Groeneveld et al., 2007a) was based on a database containing more than 31,000 data. The CHF data compilation work at the University of Ottawa was continued, and the latest reported AECL-UO data bank contains more than 36,500 points compiled from 116 datasets (Groeneveld et al., 2007b). A recent NUREG report (Groeneveld, 2017) identified 17 CHF data compilations; a description of these compilations is provided in Appendix B.

4. Evolution of CHF prediction methods

4.1. Empirical CHF correlations

CHF prediction methods for flow boiling have been reported as early as 1949 (McAdams et al., 1949). The interest in CHF initially grew slowly, but expanded rapidly in the early 1960s as an improved knowledge of CHF was urgently needed to determine the power limits for the many reactors then under construction around the world. Tong's (1965) book (the first book to cover flow boiling CHF) reviewed several of the early CHF prediction methods, the majority of which were purely empirical (Tong, 1965). In parallel, Clerici et al. (1965) published a list of over 50 CHF correlations for water-cooled tubes. Since then the number of published CHF correlations for water-cooled tubes has increased to well over 500. In contrast to pool boiling for which the early prediction methods are still in use, none of the early flow boiling CHF correlations for tubes carry much credibility today. This is partially due to the fact that the pool boiling CHF equations had a physical basis, while the tube-CHF correlations were virtually all empirical.

Most of the tube-CHF correlations show a dependency on pressure P , mass velocity G , quality at CHF X_{CHF} and tube internal diameter D and have essentially the form $CHF = f(P, G, X_{CHF}, D)$ or in a dimensionless form $CHF/(H_{fg} G) = f([p_f/p_g], [G D^{0.5}/(\sigma p)^{0.5}], [X_{CHF}], [D/D_{ref}])$ where H_{fg} is the latent heat, p_f and p_g are respectively the saturated liquid and vapor density and σ is the coolant surface tension and D_{ref} is a reference diameter (usually 8 mm) to non-dimensionalize the diameter. Other dimensionless parameters may have been employed as well. The choice of correlation parameters (P, G, X_{CHF}, D) is essentially correct, but the functional relationship between these parameters depends on the CHF mechanism which varies with flow conditions, hence the large proliferation of correlations. The CHF equation sets of Katto (1992, 1994) and Lee and Mudawar (1988) deserve special mention since they (i) have a phenomenological basis, (ii) are based on a large database, and (iii) appear to have a wider range of validity.

Although the application of empirical correlations is convenient and computationally efficient, the main drawback is their limited range of application. Frequently the reader confuses the range of the data base with the range of application of the correlation — this can even lead to negative predictions of CHF within its recommended range (e.g. Biasi Correlation in RELAP, (Groeneveld and Leung, 1988)). The large majority of the correlations can only be applied to one type of CHF and they are generally incapable of predicting the limiting quality effect.

4.2. Analytical CHF models

The lack of satisfactory CHF prediction using empirical correlations led to the development of analytical CHF models, starting in the late 1960s. These models, which were based on the physical mechanisms and satisfy the conservation equations, can be divided into two main groups:

4.2.1. Annular film dryout models

These models are based on a mass balance on the liquid film in annular flow, and postulate that CHF corresponds to the depletion of the liquid film. The original annular film dryout models were developed in the 1960s by Hewitt and co-workers (Hewitt and Hall-Taylor, 1970).

Subsequently developed models are similar but may differ in the equations for droplet entrainment and deposition and for interfacial friction and heat transfer (Whalley et al., 1974; Govan and Hewitt, 1992; Dasgupta et al., 2015; Jiao et al., 2017). These models provide reasonable predictions of CHF for the annular flow at medium to high pressures and flow rates, and at void fractions exceeding 50%.

4.2.2. Bubbly layer models

These models postulate that CHF occurs in the lower quality regime, when the bubble layer covering the heated surface becomes so thick and saturated with bubbles that no liquid mixing between the near-wall layer and the cooler core liquid is possible. Early versions of this model were proposed by Tong (1965, 1968) and Tong and Hewitt (1972). This model, and subsequent variations proposed by Weisman and Pei (1983), Ying and Weisman (1986), and Chun et al. (2000), appear to predict the CHF with a reasonable accuracy at high-pressure, high-flow, and low-quality or subcooled conditions.

Although the analytical models have improved significantly and usually predict the correct asymptotic trends after suitable fine-tuning, the evaluation process is complex and time-consuming. In the 1970s and 1980s, developing CHF models was a popular academic pursuit and as a result there are now over 50 CHF models available, each based on different assumptions of interfacial relationships. CHF models tend to be less accurate than empirical correlations over the range of the correlation's database. An excellent review of the earlier analytical CHF models has been presented by Weisman (1992).

4.3. CHF table Look-up methods

Because of the proliferation of CHF correlations and CHF models, and their limited ranges of application, the need for a more generalized CHF prediction technique is obvious. As a basis for the generalized technique, the common local-conditions-hypothesis was used, i.e., it was assumed that the CHF for a water-cooled tube and a fixed tube diameter is a unique function of local pressure, local mass flux and local thermodynamic quality. The initial attempt to construct a standard table of CHF values for a given geometry was made by Doroshchuk et al. (1975), using a limited database of 5000 data for water-cooled tubes. This table, and all subsequent tables, contain normalized CHF values for a vertical 8-mm water-cooled tube for various pressures, mass fluxes and qualities. Work on improving and expanding the CHF look-up table was subsequently undertaken at CENG-Grenoble (France), University of Ottawa (Canada), IPPE (USSR) and AECL (Canada), using an ever-increasing CHF database.

In 1986, Groeneveld et al. published the “AECL-UO look-up table”, which covers a much wider range of conditions than Doroshchuk's table. The databank on which the 1986 look-up table was based contained 15,442 CHF data points taken from 12 separate datasets and the two data compilations of Thompson and MacBeth (1964) and Zenkevich (1974). Kirillov et al. (1989a, 1989b, 1991a,b) improved the CHF table of Doroshchuk et al. by introducing a larger database, but the range of applications covered in their tables remained smaller than that of the AECL-UO table. Groeneveld and Kirillov employed different databases and methodologies to derive their respective tables. A subsequent data-exchange agreement between Groeneveld and Kirillov resulted in a combined CHF data bank of more than 30,000 tube-CHF data (24,000 data points after removing duplicates), which was used to derive the 1995 CHF look-up table (Groeneveld et al., 1996). The 1995 CHF look-up table has a wider range of validity and presents normalized CHF values for 21 pressures, 20 mass fluxes, and 23 critical qualities, covering respectively ranges of 0.1–20 MPa, 0–8 Mg m⁻² s⁻¹ (zero flow referring to pool-boiling conditions) and -0.50 to +1.00 (negative qualities referring to subcooled conditions). The 1995 look-up table also removed the sharp variations in CHF that were present at some of the boundaries between regions where experimental data were available and regions where correlations and extrapolations were employed.

These sharp variations were suppressed by employing a smoothing procedure developed by Huang and Cheng (1994).

Between 1995 and 2006, 27 additional CHF datasets for vertical tubes with upward water flow, that were not used for the development of 1995 CHF look-up table, were acquired. Further enhancements were made to the CHF look-up table and its database (such as additional screening of all CHF data sets, removal of obvious outliers using the “slice” method developed by Durmayaz et al. (2004) and incorporating the presence of a limiting quality at certain conditions), culminating in the 2006 CHF look-up table (Groeneveld et al., 2007a).

Compared to the 1995 CHF look-up table, the 2006 table (Groeneveld et al., 2007a) provides CHF values for three additional pressures (2, 4 and 21 MPa) and one additional mass flux (750 kg/m² s⁻¹). Linear interpolation between table values is used for determining CHF at in-between table conditions. Extrapolation is usually not needed as the table covers a very wide range of conditions.

Since the derivation of the 2006 CHF look-up table, additional datasets have become available. These additional datasets were reported in a recent NUREG report (Groeneveld, 2017); they can be used for future updating of the CHF look-up table or for an independent validation of the CHF look-up table.

Groeneveld et al. (1996) has compared the prediction accuracy of the CHF LUT with that of leading CHF correlations — in all cases the CHF LUT provided the better prediction accuracy. Separate assessments of the CHF LUT accuracy by Baek et al. (1997), Chandraker et al. (2000) and Durmayaz et al., (2004) have confirmed these findings.

4.4. Other CHF prediction methods

In addition to the methods described above, two other methods are being developed:

- (i). Methods based on CFD, especially two-phase CFD. Vyskocil and Macek (2010) applied CFD methods to predict CHF in a tube. Gluck (2007) applied the two-phase CFD within an advanced subchannel code to predict CHF and other flow conditions. Yadigaroglu (2014) recently reviewed the state of the art of using CMFD (Computational Multi-Fluid Dynamics) methods to predict the CHF. This approach is not sufficiently mature that it can be used with confidence in predicting CHF over a wide range of conditions (it has been used to model geometry effects in fuel bundle geometries but with limited success as far as CHF prediction is concerned).
- (ii). Methods based on ANN (Artificial Neural Networks). ANN methods have been used during the past 30 years to predict CHF. More recently a variation of this method, ANFIS (Adaptive Neuro-Fuzzy Inference System) has been applied to CHF predictions (Zaferanlouei et al., 2010). Most of the ANN-type methods use the CHF look-up tables to “train” the ANN models and as a result their accuracy can be expected to be similar.

5. Discussion

5.1. CHF data coverage

The tabulation of experimental CHF data is based on the primary CHF parameters, i.e., those parameters used in correlating the CHF, such as pressure (P), mass flux (G), quality at CHF (X_{CHF}) and diameter (D). Since the quality at CHF is a calculated parameter, it is sometimes replaced with heated length (L) and inlet subcooling (ΔH_{in}) or inlet temperature. Figs. 1–4 illustrate at which conditions CHF data have been obtained in water-cooled tubes, using respectively L/D vs. D, G vs. P, X vs. P and X vs. G plots. They include the water CHF data base used in the derivation of the 2006 CHF look-up table derivation as well as the CHF data from 27 references processed subsequent to the 2006 CHF look-up table derivation. The data density at the various flow conditions corresponds to the intensity of the grey colour of the symbols – the

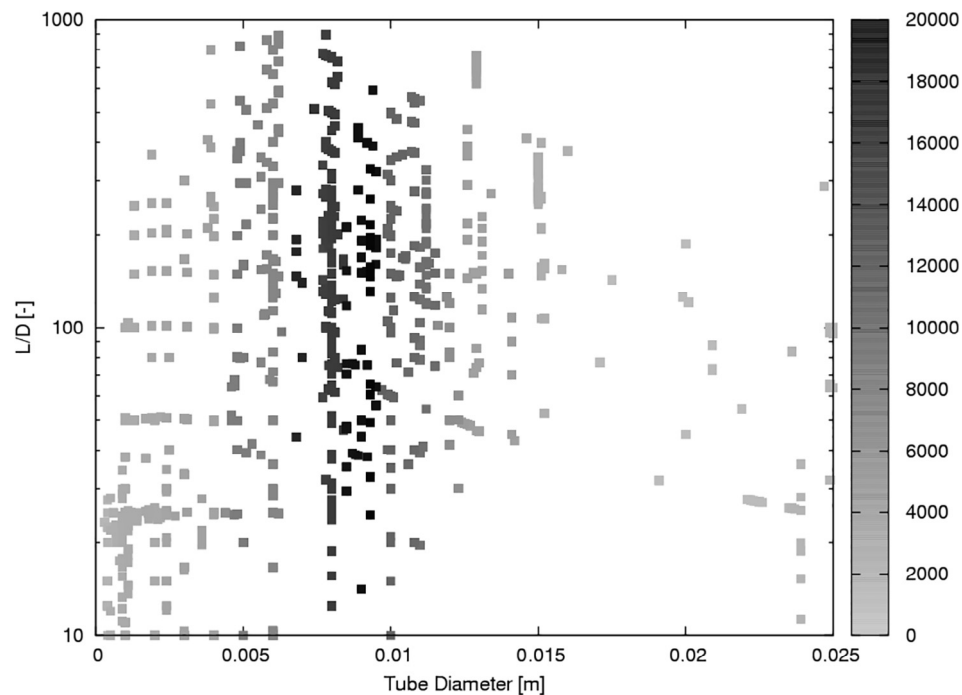


Fig. 1. CHF data coverage on an L/D vs. D map.

RHS column beside the figures quantifies the data density. The figures show that despite the large database (over 40,000 data points), there are still noticeable gaps in flow conditions and geometry where CHF data are unavailable. The primary conditions at which data are scarce or missing are:

- High flows and high qualities
- Low flows and low qualities
- Pressure range of 0.2–0.4 MPa
- The reasons for the scarcity of CHF data at the above conditions are as follows:

5.1.1. High flow/high quality

It is very difficult to obtain CHF data at high flow and high qualities since this requires the use of a complex two-phase inlet set-up. This necessitates additional heat balance calculations across a preheater or mixer to obtain the test section inlet enthalpy and inlet quality. The method of introducing the two-phase mixture into the test section can also affect the CHF. The alternative would be the use of a very long test section, which is complex and expensive and will result in a very large pressure drop. Note that, at low pressures, increasing the flow has occasionally resulted in the occurrence of critical flow conditions, which provides an upper flow limit (Tain, 1994). This is less likely to occur at

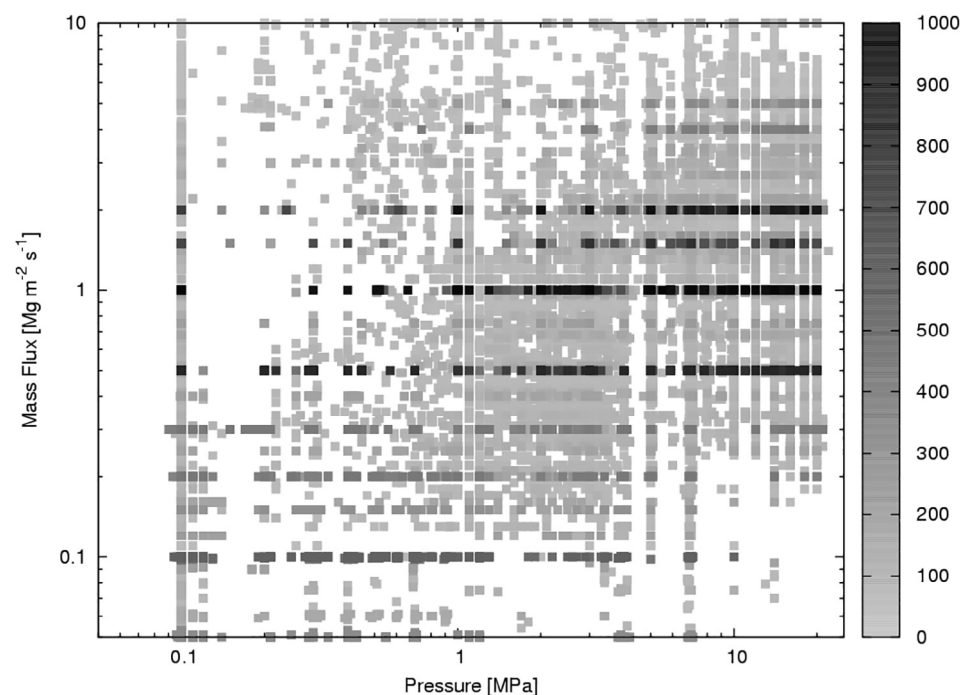


Fig. 2. CHF data coverage on a Mass Flux vs. Pressure map.

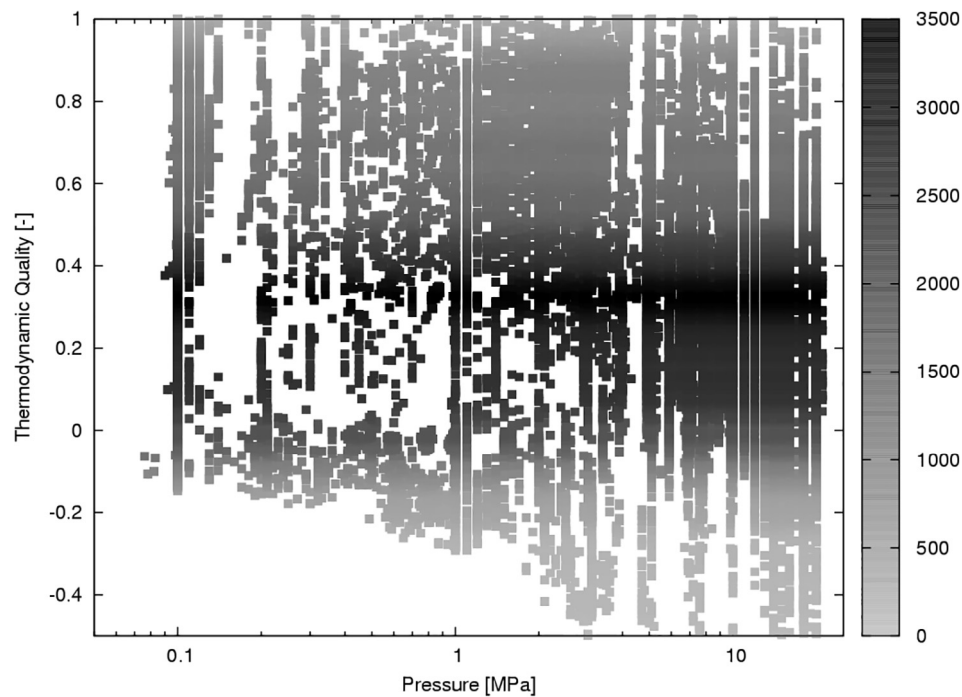


Fig. 3. CHF data coverage on a Quality vs. Pressure map.

higher pressures.

5.1.2. Low flow/subcooled CHF

At low pressures, negative qualities of less than -0.20 are impossible to obtain since they would result in water temperatures less than 0°C . In addition, the CHF is very high at these conditions and this would result in very large axial quality gradients (dx/dL). Obtaining low flow/low quality CHF data would require very short test sections, where the entrance effects could affect the CHF. In some tests at highly subcooled conditions, the CHF occurred so fast that each CHF occurrence resulted in a physical test section failure due to burnout.

5.1.3. Low pressures

Experiments at low pressure, especially for higher qualities and lower flow, can result in the occurrence of flow instability which would lower the CHF.

Another reason for the scarcity in data for some of these conditions could be a lack of interest, because heat transfer equipment may not operate at these conditions. However conditions during reactor transients can cover very wide ranges of conditions depending on the type of transient. Also current interest in Super-Critical Water Reactors increases the range of interest as the full range of pressures needs to be covered during startup and transients. Accurate prediction of CHF is

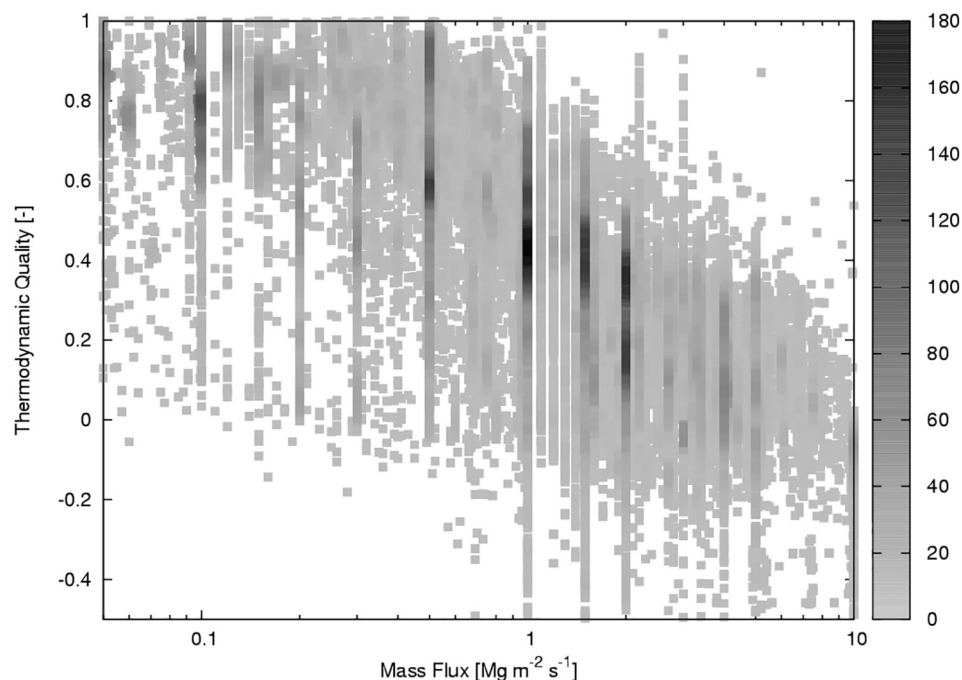


Fig. 4. CHF data coverage on a Quality vs. Mass Flux map.

important during all transients as CHF will affect the local flow conditions and heat transfer coefficients at and beyond CHF.

5.2. Uncertainty in CHF prediction

The CHF prediction uncertainties depend strongly on flow conditions although this is generally ignored. Vasic and Cheng (2000) divided the flow conditions domain in 144 subregions and assessed the availability of data sets, number of data and prediction accuracy using the CHF LUT as a reference. They found that the data dispersion as reflected by the rms error to be largest at pressures less than 8 MPa and flows less than $3 \text{ Mg m}^{-2} \text{ s}^{-1}$. To our knowledge, no assessment of the prediction error as a function of flow conditions has ever been made for empirical correlations. Empirical relationships in general do not permit to predict the impact of the limiting quality region on CHF — the largest prediction errors are expected to occur at qualities at and beyond the limiting quality. In all cases the prediction uncertainty cannot be assessed for conditions beyond the data base — the greater the extrapolation beyond the data base, the higher the uncertainty.

5.3. Dataset listing

Groeneveld (2017) has compiled references to all 160 known sources for CHF experimental data obtained in water-cooled tubes. The flow conditions of many of the datasets are also provided in this reference.

6. Conclusions and final remarks

- During the past 70 years, CHF measurements in water-cooled tubes have been obtained in over 160 separate experimental studies. Most of the early experiments lack sufficient information to assess the uncertainty in the data.
- The uncertainty in the measured CHF, based on values obtained in similar test section at different laboratories, is generally less than 10%.
- Despite the large database (over 40,000 data points), there are still noticeable gaps in the database at flow conditions where CHF data are unavailable. The primary conditions where data are scarce or missing are: (i) high flows and high qualities, (ii) low flows and low qualities, and (iii) pressure range of 0.2–0.5 MPa.
- Because of the proliferation of CHF correlations (> 500) and CHF models (> 50), and the limited range of application of the models and correlations, there was an urgent need for a more generalized CHF prediction technique. The CHF look-up table was developed in response to this need.
- The CHF look-up table is basically a normalized databank. Compared to other available prediction methods, the look-up table approach has the following advantages: (i) greater accuracy, (ii) wider range of application, and (iii) correct asymptotic trend.
- The uncertainty in the regions where data are scarce or non-existent depends on the level of extrapolation from data-based regions. It is expected to be small at conditions slightly beyond the range of data but becomes larger as the extrapolation is further beyond this range.
- Since the derivation of the 2006 CHF look-up table, additional datasets have become available; 107 references to the additional datasets were provided by Groeneveld (2017). These additional datasets can be used for future updating of the CHF look-up table or for an independent validation of the CHF look-up table.
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Appendix A. Reproducibility studies

A.1. CISE (1970) and Nilsson (1970) European CHF reproducibility exercise

Because of concerns that CHF measurements could vary significantly between different laboratories, even when the test equipment and the test conditions are nominally identical, the European Two-Phase Group decided to perform a so-called “CHF reproducibility” exercise. The objective of this reproducibility exercise was to determine the variation in CHF measurements between various laboratories. The results of this exercise are reported by two sources: CISE (1970) and Nilsson (1970). The CISE report is very large, containing individual chapters written by each of the participating laboratories. Nilsson's report also contains all individual datasets (a total of 594 CHF measurements were taken by the participating laboratories) and describes the analysis of the data.

The following organizations/laboratories participated in the exercise:

- AE, Sweden
- AEG-Telefunken, Germany
- CEA, France
- CISE, Italy
- EUR, Ispra, Italy
- MAN, Germany
- Sorin, Italy
- UKAEA, United Kingdom

The nominal test conditions were:

Tube ID: 10 mm
 Tube Heated Length: 2 m
 Pressure: 3, 5, 7, and 9 MPa
 Mass Flux: 260 to $6000 \text{ kg m}^{-2} \text{ s}^{-1}$
 Inlet Subcooling: 15 to 3235 kJ/kg (depending on pressure)

Note that the above ranges of conditions were the maximum ranges; the ranges of conditions were different for the various laboratories.

The discrepancy in CHF values was expected to be within 10%; however, the discrepancy initially exceeded 30%. After careful examination of the two outliers (the other ones were within 10%), two laboratories found inconsistencies in their measurement approach and hardware (which may have given rise to flow instabilities) and one laboratory withdrew its results.

A.2. Kirillov (1984, 1985, 1997): USSR CHF Reproducibility Study

Kirillov et al. (1984, 1985) compared the results of a unique collaborative CHF experimental investigation performed at 10 different laboratories in the USSR. In total, 2470 CHF measurements were taken. The objective of this collaboration was to compare the CHF obtained in various experimental test facilities at the same nominal conditions.

All CHF experiments were performed in tubes of Kh18NiOT steel (stainless steel with 18% Chrome and 10% Nickel) having an ID of 8 mm and a wall thickness of 2 mm. The tube lengths were 1, 3 and 6 m. All 10 participating laboratories used tubes produced from the same batch. The test conditions were:

Pressure: 6.37–18.04 MPa
 Mass Flux: 494–4154 $\text{kg m}^{-2} \text{ s}^{-1}$
 Quality at CHF: -0.494 to 0.981

Inlet Subcooling: $7\text{--}1537 \text{ kJ kg}^{-1}$
 Heat Flux at CHF: $110\text{--}7700 \text{ kW m}^{-2}$

This CHF reproducibility exercise was performed during 1981–1983 and yielded ~2500 experimental values of the CHF power. No precise references for the various laboratories and their results were given; Kirillov referred to them as:

- VTI
- F. E. Dzerzhinskii All-Union Heat Engineering Institute (FEI)
- Polzunov Central Boiler-Turbine Institute (TsKTI)
- V. Kurchatov Institute of Atomic Energy (IAE),
- Institute of Engineering (or Technical?) Thermophysics of the Academy of Sciences of the Ukrainian SSR (ITTF)
- Special Design Office (SDO) “Gidropress”
- NIKIET
- Scientific-Production Union (SPU) “Energiya”
- Kiev Polytechnical Institute (KPI), Ukraine
- Elektrogorsk Research Station (ENIS)

The results showed that the spread in CHF power was 16 to 32% for the 1 m heated length and 10% for the 3 m heated length. The spread in CHF on a CHF vs. Critical Quality plot was significantly greater (this was expected as the uncertainty in CHF based on a constant inlet temperature are always smaller than the uncertainty in CHF based on a constant calculated dryout quality (Groeneveld, 1996)). Very little additional information was provided by Kirillov.

Appendix B. Description of CHF data compilations (listed chronologically)

B.1. DeBortoli et al. (1958)

The earliest compilation of CHF data is that of DeBortoli et al. (1958), who collected over 1000 tube-CHF data. Included in this report are also new experimental data obtained by DeBortoli and co-workers in tubes and rectangular channels. CHF measurements were obtained in a directly heated tube with an ID of 0.187 in. (4.75 mm) and a heated length of 12.5 in. (0.318 m). The test section material was “A” nickel. Burnout detection was with a thermocouple attached to the test section near the outlet end of the heated length. Only a partial pdf file for this report was available (the original was lost). However, the database by DeBortoli et al. was also provided by Firstenberg et al. (1960). Thompson and MacBeth (1964), in their CHF data compilation, also list 686 of the DeBortoli et al. 1958 data obtained in tubes.

B.2. Firstenberg et al. (1960)

As a first step in arriving at a better correlation of burnout heat fluxes for the conditions of interest to boiling water reactors, Firstenberg et al. collected experimental CHF data, obtained in the net-quality region, from the literature. This compilation is an updated extension to the earlier compilation by DeBortoli et al. (1958), and contains an additional 414 CHF data points obtained in tubes. Note that some of the data are obtained in rectangular channels.

B.3. Thompson and MacBeth (1964)

One of the earliest significant compilations of tube-CHF data was that of Thompson and MacBeth (1964). They compiled 4389 data taken from 12 datasets. In compiling their data, they ignored any data taken with a two-phase inlet condition as this could have affected the CHF values.

B.4. Becker et al. (1965)

Becker et al. published all Swedish tube-CHF data obtained up to 1965, and included selected datasets from the UK and USA. These data are listed in two reports: AE-177 (Becker et al., 1965), and AE-178 (Becker, 1965).

Table I of AE-177 contains 3473 Swedish tube-CHF data, while Table I of AE-178 contains 1260 Swedish tube-CHF data. In addition, Tables II and III of AE-178 contain about 1000 tube-CHF data obtained elsewhere.

B.5. Zenkevich (1974)

This translation is a very large report of more than 400 pages, containing about 7000 CHF data points, many of them obtained in uniformly heated tubes of which about 650 were obtained by Zenkevich. This is a valuable resource that provides an excellent summary of data obtained in the USSR and elsewhere. Some additional datasets obtained by seven different authors are also included. The odd-numbered tables between Tables 16 and 35 of this reference contain CHF data obtained in tubes but with a non-uniform axial flux distribution; these should therefore be ignored. Table 1 of this report provides the ranges of conditions covered by 31 pre-1974 CHF data known by Zenkevich, some of which had a two-phase inlet condition.

B.6. Doroshchuk et al. (1975)

Doroshchuk et al. (1975) reported having constructed a database of around 5000 CHF tube data obtained primarily in the USSR. This database was subsequently used for constructing the first CHF look-up table but covered only a limited range of flow conditions.

B.7. Groeneveld et al. (1986)

Groeneveld et al. (1986) lists references to 12 individual datasets and two data compilations (Thompson and MacBeth, 1964; Zenkevich, 1974), containing in total 15,442 CHF data used for deriving the 1986 CHF look-up table. This database has been referred to elsewhere as the “AECL-UO CHF database”, as it was based on a cooperative effort at the University of Ottawa (UO) and Atomic Energy of Canada Ltd. (AECL).

B.8. Kirillov (1992)

Kirillov and co-workers at the Institute of Physics and Power Engineering (IPPE) in Obninsk compiled a CHF database of over 14,000 points from primarily USSR sources. These data were transferred to the University of Ottawa around 1992. Subsequently, Groeneveld and Kirillov combined their databases expanding the total number of data points beyond 29,000. The combined database is referred as the “AECL-UO-IPPE CHF database” and this was the basis of the 1995 CHF look-up table (Groeneveld et al., 1996).

The database transferred to Groeneveld by Kirillov consisted of six datasets that were identified by the original experimenter. The experimenters and the number of data in each subset of Kirillov's database are listed below:

- Alekseev et al. (1964) 1108
- Zenkevich et al. (1969) 5641
- Zenkevich et al. (1971) 392
- Ladislau (1978) 136
- Smolin et al. (1962) 3009
- Bergel'son (1980) 336
- Kirillov et al. (1984) 2470

The full reference of these datasets is not available. Note that some of the data in this database contributed to the reproducibility exercise

described by Kirillov et al. (1985) and Kirillov (1997).

B.9. Vandervort (1992)

Vandervort compiled subcooled CHF data obtained in small diameter tubes and with high to very high mass fluxes, resulting in very high CHF values. The ranges of CHF conditions of his database were:

Mass Flux: 3000–90,000 kg m⁻² s⁻¹
 Subcooling: 0–200 K
 Pressure: 0.1–3.0 MPa
 Diameter: 0.3–3.0 mm
 L/D Ratio: 1.0–40.0
 Heat Flux at CHF: 6 * 10⁶ to 2 * 10⁸ W/m²

The CHF database contains seven datasets, containing over 800 CHF data points taken from the references listed below:

- Ornatskii (1960)
- Ornatskii and Kichigan (1962)
- Ornatskii and Vinyarskii (1964)
- Loosmore and Skinner (1965)
- Nariai et al. (1987)
- Boyd (1989)
- Vandervort (1992)

B.10. Celata and Mariani (1992)

Celata and Mariani's database contains 1887 CHF data from 23 references; they were scanned and digitized in the 1990s. Of these data, 1357 were obtained in round tubes and were tabulated separately. This database was transferred to the University of Ottawa by G.P. Celata (see reference to Celata (1993)). The data are from Celata et al. (1993), and from Reference 4 in Celata and Mariani (1992) which is an unpublished ENEA report no longer available.

B.11. Soderquist (1994)

The electronic copy of this large dataset was obtained directly from the author, Bo Soderquist from the Department of Nuclear Reactor Engineering, Royal Institute of Technology, Stockholm, Sweden. The electronic version contains 1485 data points. Subsequently, a printed copy of the data and the experimental details were received as well.

B.12. Groeneveld et al., (1996)

Groeneveld et al. (1996) significantly expanded the 1986 AECL-UO database reported by Groeneveld et al. (1986). Aside from including many additional datasets, the 1986 AECL-UO CHF database also included USSR data obtained through a data exchange agreement between the University of Ottawa and IPPE (USSR). This resulted in a combined CHF data bank of more than 30,000 tube-CHF data (24,000 data points after removing duplicates). This combined database, referred to as the "AECL-UO-IPPE CHF database", was used to derive the 1995 CHF look-up table (Groeneveld et al., 1996).

B.13. Hall (1999), and Hall and Mudawar (2000)

Hall (1999) and Hall and Mudawar (2000) compiled a database of over 30,000 tube-CHF data, which covered tubes of very small diameters ($D < 2$ mm). The Purdue University Boiling and Two-Phase Flow Laboratory (PU-BTPFL) CHF database for water flow in a uniformly heated tube was compiled from the world literature dating back to 1949 and represents one of the largest CHF database ever assembled with 32,544 data points from over 100 sources. It includes flow in horizontal channels and downflow as well. Although Hall and

Mudawar's database was not available for the derivation of the 2006 look-up table (Groeneveld et al., 2007a), almost all their data sources were known to the authors.

B.14. Groeneveld (2017)

The AECL-UO 1996 database was expanded significantly between 1995 and 2006 as 27 additional CHF datasets for vertical tubes with upward water flow were acquired by the University of Ottawa. The combined database contained 31,000 data points from 61 datasets. After the data were thoroughly screened (by removing duplicates, outliers, etc.), the enlarged databank was normalized resulting in the 2006 CHF look-up table (Groeneveld et al., 2007a). A discussion of the screening process has been described in detail by Groeneveld et al. (2007b).

B.15. Groeneveld (2017)

The most recent review of all available CHF data obtained in water-cooled tubes was sponsored by the USNRC and has been reported in a recent NUREG report (Groeneveld, 2017). This study identified the 17 separate tube-CHF data compilations and over 160 CHF datasets, many of which were not processed or used in deriving the CHF look-up tables. As shown in Figs. 1–4, the combined database now covers a wider range of flow conditions than the database used for deriving the 2006 CHF look-up table. This implies that an update of the 2006 CHF look-up table by normalizing this expanded database would improve the accuracy and validity range of the look-up table.

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